UAiTED
University Alliance in Talent Education Development

# 2025 UAiTED Innovation Competition

## Proposal

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| **Group Name** | Meownovators |
| **Selected Theme** | Green Building/ Urban Solutions |
| **Project Title** | EcoFusion Septic |

**Abstract of the project**

Urban sewage management is an ongoing crisis that has the potential to greatly impact the sustainability of our environment and urban city development. By statistics, over 80% of wastewater is discharged into the world without suitable treatment (UNESCO, 2017). To make matters worse, in most developing countries such as Indonesia or Malaysia, the value can hit up to 90%. Taking the case study nation, Malaysia, as an example, around 30% of the population live outside a centralized sewage system and are heavily reliant on aging or poorly managed septic tanks (Suruhanjaya Perkhidmatan Air Negara, 2023). Such inefficiency contributes to severe water pollution, with the status of rivers being classified as polluted increasing from 28% in 2015 to 45% in 2022 (Department of Environment Malaysia, 2022). As urban populations grow, these problems intensify and exert immense pressure on both water bodies and public sanitation systems.

One of the consequences of improper sewage management is water pollution, which potentially causes both environmental and health risks. The contaminated water, along with the solid waste within, contains dangerous pathogens such as *E. coli* or even *cholera*. This will result in widespread waterborne diseases. According to WHO, up to 829 thousand people die annually from diarrhea-related diseases due to unsafe water and sanitation. For Malaysia, up to 50% of the reported communicable diseases are directly linked to water pollution, of which one of the main contributors is improper sewage management. Moreover, untreated sewage often carries residues such as nitrogen compounds and phosphorus compounds, which affect the water quality of aquatic life. Hence, our design aims to eliminate water pollution by creating a more effective water treatment plant that converts the solid waste into bioplastic while allowing the liquid and gaseous waste to be converted into electricity.

Furthermore, the economic impact is detrimental to the economy. Malaysia's economy loses several millions of money annually due to declining water quality, leading to reduced tourism and increased treatment cost (National Water Services Commission, 2023). Furthermore, urban flooding due to clogged drainage from improper sewage disposal has resulted in cities such as Kuala Lumpur losing over 100 million ringgit annually (The Star, 2022). The insufficiency in handling human waste also results in a loss of opportunity for resource discovery, as the current status quo has the tendency to discard waste by landfilling rather than utilizing it for energy or materials. Hence, our design aims to utilize solid waste into bioplastic and generate energy from liquid and gaseous waste.

Poorly managed septic tanks, if not taken care of regularly, leak waste into the soil and groundwater, contaminating nearby drinking water resources. This pragmatic situation worsens in urban and suburban areas where space and infrastructure are highly limited and the population is dense, which has a higher risk of resulting in a disease outbreak and groundwater pollution. Our design can resolve the problem by reducing and reusing the waste, hence preventing leakage from the root. Lastly, the presence of greenhouse gases emitted during the decomposition of sewage wastewater—methane, to be specific—contributes significantly to climate change, where wastewater is responsible for up to 1.57% of global emissions (IPCC, 2019). Hence, our design utilizes these greenhouse gases into renewable energy.

The EcoFusion Septic offers a sustainable solution to the escalating sewage crisis by turning harmful waste into valuable resources. Traditional septic systems often leak untreated waste into water bodies, causing diseases like cholera and hepatitis A. Our design tackles this by converting solid waste into bioplastics using engineered E. coli and Halomonas sp., reducing environmental and health hazards while creating a biodegradable alternative to petroleum-based plastic. Simultaneously, methane gas from liquid waste is captured and used to generate energy through *Methanosarcina barkeri*, supplying up to 260 MWh/day—enough to power a small city. This closed-loop system prevents pollution, lowers emissions, and operates independently of central sewage infrastructure, making it ideal for rural and urban use. By treating waste as a resource, our system provides a clean, scalable, and circular approach to modern sanitation for Green Building and Urban Solutions

**Outline of the solution**

Overview

The EcoFusion Septic consists of two main components: power generation and bioplastic synthesis. The power generation component utilizes gaseous waste and converts it into electricity, while the bioplastic synthesis component converts solid waste into bioplastic. This design aims to maximize the efficiency of the SMART Septic Tank.

Bioplastic Synthesis

The synthesis of bioplastic will be done by utilizing the solid waste. Solid waste, also known as the Sludge Layer (Tank, A-1.S. 2023), mainly consists of organic matter such as feces and food waste (US EPA, OW 2019). In this proposal, we assume that all organic matter is of the same type of waste—feces—for easier calculation and proving purposes. The nature that food waste shares an almost similar content of composition in comparison to feces supports this claim (Appendix 1). The process can be segregated into three steps: fermentation, conversion, and extraction, assuming that all processes are done under standard atmospheric pressure.

First step, fermentation. The solid waste is converted into Volatile Fatty Acid, or VFA in short. We will be using a genetically modified bacterium, *Escherichia coli*, or *E. coli* for short, created by the University of Calgary. The *E. coli* is modified in several genetic codes such as *phaJ*, *phaC*, and *phaCBA* genes from bacteria such as *Ralstonia eutropha* and *Pseudomonas aeruginosa* to convert carbon compounds into VFA (Vamshi, R., Noorbakhsh, J., Rosales, A., Rajagopal, R., Tang, M., Wu, M. and Xu, Y., 2017). The conversion rate of solid waste into VFA is around 82% (Appendix 2). Given that around 200g of solid waste is produced per day, we can estimate up to 20.5 tons of VFA can be produced per day (Appendix 3).

The second step of the process is to convert the fermented VFA into a usable bioplastic monomer. Plastic is a polymer that is created via synthesis of a large monomer chain (Rodriguez, F. 2018). In this case, the monomer we are using will be *Polyhydroxyalkanoates*, or PHA in short. The VFA is fed to the *bacterium Halomonas sp.* at a temperature of 30 degrees Celsius with an acidity level of 7 pH in an industrial centrifuge, when the VFA solution is then added into it along with 20% of Soytone. The centrifuge will be operating at 90 rpm and the VFA is fed again at the 26th hour, 46th hour, and 56th hour. The entire process will last for 90 hours (Park, Y., Jeon, J.-M., Park, J.-K., Yang, Y.-H., Shin Sik Choi and Yoon, J.-J. 2023). At the end of the process, a yield of 73% is obtained, which is around 14.965 tons of PHA monomer being produced (Appendix 4).

The last step in the bioplastic synthesis is extraction. The biomass containing PHA is suspended in sodium hypochlorite and incubated for 1–2 hours in a 37-degree Celsius environment. The liquid is then centrifuged to form a PHP pellet that is washed with methanol or ethanol and dried for further usage. The yield is around 50%; hence, our final yield of bioplastic per day is 7.4825 tons (Appendix 5).

Power Generation

The gaseous waste collected in septic tanks is methane gas and other biogases (Organica Biotech, 2025). These biogases have a relatively high energy generation rate of 43.5% (CAT, 2025). The organic waste includes human feces, kitchen wastewater, and other biodegradable material.

In this case, the solid waste is first isolated via sedimentation and other physical filtration methods (EPA, 2025). The leftover liquid and gaseous waste is then conducted to the Generator Chamber, where fermentation, gas collection, and energy generation are processed. The liquid is first fermented using Methanosarcina barkeri, which can generate a relatively high concentration (De Vrieze, J., Hennebel, T., Boon, N., and Verstraete, W., 2012). The gas collected per person per day is approximately 0.5 $m^3$, with a rough estimate of 250g of solid feces per day per person (Rose, C., Parker, A., Jefferson, B., and Cartmell, E., 2015), which can generate up to 0.1359 kilowatts per day using the generator CG132B-16 (CAT, 2025). The extraction of bioplastic will result in an energy consumption of around 5.95 MWh per day (Appendix 6). Assuming a city in Malaysia with up to 1,982,112 residents, daily energy generation would reach up to 260 MWh per day, which is equivalent to the overall consumption of Cambridge, UK (Templar.co.uk, 2016).

**Business Strength and Impact**

At the heart of EcoFusion Septic's value proposition is its Triple Revenue Stream Architecture, with the primary source of income derived from the production and sale of premium biodegradable polyhydroxyalkanoates (PHA). These bioplastics, valued between $3,000 to $4,000 per ton, represent a lucrative opportunity in a rapidly expanding global market (Polyhydroxyalkanoates (PHA) Price Trend, Index, Demand, n.d.). The PHA market is also projected to reach USD 195 million by 2028, growing at a robust CAGR of 15.9%. (MarketsandMarkets, 2025). This growth trajectory is underpinned by a global shift away from petroleum-based plastics due to increasingly stringent government regulations and a growing consumer preference for sustainable materials. PHAs, being fully biodegradable and derived from renewable biological sources, are gaining traction in industries such as packaging and food services, biomedical applications, and agriculture. Their eco-friendly profile and versatility position them as a strategic alternative to traditional plastics. Given our initial expectation of 7.4825 production of bioplastic daily, we are able to generate an annual revenue between $8.2 and $10.9 million.

A key differentiator lies in the platform's proprietary biotechnology. The system employs genetically modified E. coli with enhanced conversion efficiencies, Halomonas species with optimized PHA yields and Methanosarcina barkeri for efficient methane capture and energy generation. These integrated biotechnological components form a complex multi-stage system that presents substantial technical and regulatory barriers to entry. The use of genetically modified organisms (GMOs) introduces strict biosafety and environmental compliance requirements, often resulting in approval timelines of 18-24 months, particularly in markets governed by precautionary principles.

Beyond economics, EcoFusion Septic delivers profound public health and environmental benefits, addressing systemic challenges faced by rapidly urbanizing regions. As mentioned before, approximately 829,000 deaths annually are caused by diarrheal diseases linked to unsafe water, sanitation, and hygiene, with more than half of all communicable diseases in some developing countries traced back to contaminated water sources. By significantly improving wastewater treatment and preventing the spread of waterborne pathogens, EcoFusion Septic has the potential to reduce disease burdens and yield healthcare savings of around $1,200 per prevented case, as estimated in economic studies on the cost of illness and healthcare interventions (Prüss-Üstün et al., 2016).

The system also prevents sewage contamination of groundwater, a critical concern in dense urban environments where aquifers serve as essential drinking water sources. Groundwater remediation projects are notoriously expensive, with costs ranging from $2 to $10 million per site, depending on the level of contamination and geography (U.S. Environmental Protection Agency, 2020). By intercepting waste before it enters water tables, EcoFusion Septic eliminates these costs and protects public water supplies.

The platform's climate benefits are equally notable. Methane has 84 times more global warming potential than $CO_2$ over a 20-year period, and wastewater is a major source of fugitive methane emissions. Each EcoFusion Septic deployment captures enough methane to equate to removing approximately 500,000 cars from the road annually, based on EPA equivalency calculations (U.S. EPA, 2023). In addition, by producing biodegradable bioplastics, the system displaces the need for roughly 2,730 tons of petroleum-based plastics per city-scale installation, reducing reliance on fossil fuels and mitigating plastic pollution.

**Novelty**

In contrast to ordinary septic systems which store or landfill the waste, our design converts waste into bioplastic. Allowing both circular economy principles and green technology to be applied in our design at micro-infrastructure level. Such design is rarely, if ever, implemented in household level sanitation systems (World Bank, 2020).The concept of energy-positive sanitation is novel in itself; most waste systems are energy-intensive and rely on external grids. EcoFusion Septic is not only self-sustaining by itself, it can also provide energy generation from feces, making it a scalable model (UNEP, 2021). This itself is a revolution in Green Building and Urban development

Additionally, most bioplastic is synthesized from agricultural waste or food waste when synthesizing bioplastic from urban wastewater is a novice idea in applied systems (OECD, 2018). The immense potential of generating PHA directly from urban fecal waste makes our design a first of its kind to combine environmental sanitation with bio manufacturing makes it a perfect Green Building

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## Appendix

Appendix 1: Comparison of Feces composition to Food Waste composition

| Component | Feces | Food Waste |
|---|---|---|
| **Carbohydrates** | 25–50% (incl. undigested fiber) | 40–60% (starch, sugars, fiber) |
| **Proteins** | 20–30% | 10–20% |

| | | |
|---|---|---|
| **Fats** | 10–20% | 10–30% |
| **Ash/Minerals** | 10–15% | 5–10% |

Table 1A: Comparison of Feces composition to Food Waste composition

Reference: Jimenez, B. and Maya, C. (2014), Li, R., Wang, J., Zhang, Z. and Shah, A.N. (2013),Rose, C., Parker, A., Jefferson, B. and Cartmell, E. (2015), Zhang, D., Chen, Y., Zhao, Y. and Zhu, X. (2010)

Appendix 2:Theoritical Yield of VHA

123g of VFA can be generated from 150g of feces

123/150=82% yield

Appendix 3: Mass of VHA collected Per day

1982112 individuals

Average 250g=0.25kg of feces

495528 x 82%=406332.96kg=406 tons

Since our centrifugal can only support 25 tons, we will only use 25 tons as our reactant

Appendix 4: Yield of PHA Per day

20.5 tons x 73%  14.965tons

Appendix 5: Final Yield of Plastic Per day

14.965 tons x 50% = 7.4825

Appendix 6: Energy Consumption of Centrifuge

$$Energy\ consumption = Power\ usage \times Time\ (hours)$$

$$Energy\ consumption = 255\ kW \times 210\ hour$$

$$Energy\ consumption = 53.55\ MW$$

$$Energy\ consumption\ per\ day = \frac{53.33\ MW}{9\ day}$$

$$Energy\ consumption\ per\ day = 5.95\ MW/day$$

Reference: (ANDRITZ, n.d.)

Appendix 7:Case Study Of The Amount Of Electrical Energy Produced by Faeces Per Day In Wilayah Persekutuan Kuala Lumpur

*7.1 Population of Wilayah Persekutuan Kuala Lumpur*

Current population of Wilayah Persekutuan Kuala Lumpur is 1982112 people (Department of Statistics Malaysia, n.d.).

*7.2 Human Stool Parameter*

| | Parameter of Human Stool | Value |
|---|---|---|
| | Wet mass | 250 g/human/day |
| | Composition | 75 % of water and 25 % of solids |

Table 7A (Metcalf & Eddy, 2014); (Rose et al., 2015); (WHO, 2018).

*7.3 Methane production from faeces*

| | Types of faeces | Amount of methane production (under anaerobic digestion) |
|---|---|---|
| | Dry solids faeces | 0.5 $m^3$ $CH_4$ per kg of faeces |

Table 7B (Rose et al., 2015); (Metcalf & Eddy, 2014); (Parkinson et al., 2014).

*7.4 Turns amount of wet faeces to amount of solid faeces*

250 g wet faeces = 250 g ×0.25 solid faeces

=62.5 g solid faeces

=0.0625 kg solid faeces

*7.5 Production of Methane*

Production of CH4 =0.5 m3 CH4/1kg solid faeces ×0.0625 kg solid faeces

=0.03125 m3 CH4

*7.6 Thermal Energy Production By Methane*

| No. | Methane, $CH_4$ | Thermal Energy |
|---|---|---|
| 1 | 1 $m^3$ | 10 kWh |

Table 7C (IPCC, 2006); (FAO, 2008).

*7.7 Thermal energy produced by 0.03125 m3 methane*

0.03125 m3 CH4 ×10 kWh/m3 CH4 =0.3125 kWh

*7.8 Electrical conversion efficiency for large-scale biogas generations*

Electrical conversion efficiency for large-scale biogas generations is 43.5 % (Cat® Gas Gensets Help Convert Biogas to Clean Energy, n.d.).

*7.9 Amount of electrical energy generated per day*

0.3125 kWh ×0.435=0.1359375 kWh/day

*7.10 Amount of electrical energy generated by 1982112 human's faeces per day*

0.1359375 kWh ×1982112 human =269443.35 kWh/day